

1 Checking Convexity Intepolation of points

Questions: Given a bunch of points $(x_i, y_i)_{i=1, \dots, n}$, with $x_i \in \mathbb{R}^d, y_i \in \mathbb{R}$, can this interpolate a convex function?

Idea: Points (x_i, y_i) can interpolate a convex function means that there exist a convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $f(x_i) = y_i$. We have such convex function if and only if for all $i = 1, \dots, n$, there exist subgradient vector $g_i \in \mathbb{R}^d$ such that for all $j = 1, \dots, n$

$$y_j \geq y_i + g_i^T (x_j - x_i)$$

This means for every point (x_i, y_i) , there is a supporting hyperplane that passes through (x_i, y_i) and keeps all other points on or above it. (Prove: the necessary condition already follows from definition of that convex function. For sufficient condition: we can build a convex function $f(x) = \max_{i=1, \dots, n} \{y_i + g_i^T (x - x_i)\}$, and $f(x_i) = y_i$ since we have for all $k \neq i, y_k + g_k^T (x_i - x_k) \leq y_i$.)

Now using this idea, we can construct a linear program: For $i, j = 1, \dots, n$,

$$y_j \geq y_i + g_i^T (x_j - x_i)$$

where $x_i, x_j \in \mathbb{R}^d, y_i, y_j \in \mathbb{R}$ are known vectors and values. And only g_i 's are unknown vectors. If this linear program is feasible, then we know there exist such subgradient vectors, so points can interpolate a convex function. If the linear program is not feasible, then points can not interpolate convex functions.

In the original problem, there are totally N points to check, and there are N inequalities for each point. So the computational cost would be high if we have large amount of points. To deal with that, we can check the dual form of inequalities for each points. We can do that by using Farkas' lemma:

Either $Ax \leq b$ has a solution $x \in \mathbb{R}^n$, or $A^T y = 0$ has a solution $y \geq 0$ with $b^T y < 0$.

Apply Farkas Lemma to our original inequalities for each point i , we have $y_j \geq y_i + g_i^T (x_j - x_i)$ is not feasible if and only if $\sum_j \alpha_j (x_j - x_i) = 0$ has a solution $\alpha \in \mathbb{R}_+^n$ with $\sum_j \alpha_j (y_j - y_i) < 0$. Since CVXPY does not take strict inequality, we instead make a LP formation:

$$\begin{aligned} & \min_{\alpha \geq 0} \sum_j \alpha_j (y_j - y_i) \\ & \text{s.t. } \sum_j \alpha_j (x_j - x_i) = 0 \end{aligned}$$

We check the LP is unbounded or not. If it is unbounded, then it means that $\sum_j \alpha_j (x_j - x_i) = 0$ has a solution, and then our original inequality is infeasible. Vice versa.

We have implemented an LP blackbox for dealing this problem. There are several places to improve: 1. using different formulation of LP. 2. using different packages. 3. using parallelism to check the LP for all points at once.

2 Adaptable Step Size via Convexity Check Using LP

In the original implementation of Tabak and Turner, each update to the potential function was carried out with a fixed coefficient β multiplying the chosen basis function $F_{k+1}(z)$. Specifically, after k steps the

updated potential had the form

$$u_{\beta}^{k+1}(z) = u_{\beta}^k(z) + \beta_{k+1}F_{k+1}(z),$$

where both u_{β}^k and F_{k+1} are convex functions.

If $\beta_{k+1} \geq 0$, convexity of the updated potential is automatically preserved. Allowing $\beta_{k+1} < 0$ can improve flexibility and fitting power, but convexity of u_{β}^{k+1} is no longer guaranteed.

2.1 Motivation

To ensure the transport map remains the gradient of a globally convex potential on the data domain, we empirically test convexity on a fixed sample cloud $\{z_0^{(i)}\}_{i=1}^m$. We check the finite-family of affine minorant inequalities: for every point i , there exists subgradient vector $g_i \in \mathbb{R}^n$ such that

$$u_{\beta}^{k+1}(z_0^{(j)}) \geq u_{\beta}^{k+1}(z_0^{(i)}) + g_i^{\top}(z_0^{(j)} - z_0^{(i)})$$

for all points j . This is equivalent to the convex interpolation problem on the sample set. As we fix a collection of sample points $\{z_0^{(i)}\}_{i=1}^m \subset \mathbb{R}^n$ drawn once at initialization. At each iteration we evaluate the updated potential at these points:

$$(z_0^{(i)}, u_{\beta}^{k+1}(z_0^{(i)}))_{i=1}^m.$$

and we check if these points can interpolate a convex function using the LP blackbox.

2.2 Adaptable β Rule

- If the LP is feasible, we accept the candidate β_{k+1} .
- If the LP is infeasible, we reduce the absolute value of β_{k+1} (e.g. via multiplying a constant factor) until feasibility is restored.

This ensures that each update preserves empirical convexity relative to the chosen sample points, preventing invalid potentials while maintaining flexibility.

Algorithm.

- (1) Set backtracking limit $B=20$, shrink factor $s=1.25$, and an initial candidate $\beta^{\text{cand}} < 0$.
- (2) **For** $t = 1, \dots, B$:
 - (1) Define $u_{\beta}(z) = u^k(z) + \beta^{\text{cand}}F_{k+1}(z)$.
 - (2) Let $x_i \leftarrow z_0^{(i)}$ and $y_i \leftarrow u_{\beta}(x_i)$ for $i = 1, \dots, m$.
 - (3) Run the LP convex interpolation check on $\{(x_i, y_i)\}_{i=1}^m$.
 - (4) **If** feasible, accept $\beta_{k+1} \leftarrow \beta^{\text{cand}}$ and **stop**.
 - (5) **Else** set $\beta^{\text{cand}} \leftarrow \beta^{\text{cand}}/1.25$ and continue.

2.3 Some Experiment Results

The implementation details, together with reproducible code and example notebooks, are available on GitHub:

[https://github.com/echen347/polymath25-ML/blob/main/Normalizing%20Flows/Normalizing_flow_and_convexity_check%20\(updated%20with%20Jacobian%20and%20inverse%20map%20computation%20and%20convexity%20check%20using%20LP\).ipynb](https://github.com/echen347/polymath25-ML/blob/main/Normalizing%20Flows/Normalizing_flow_and_convexity_check%20(updated%20with%20Jacobian%20and%20inverse%20map%20computation%20and%20convexity%20check%20using%20LP).ipynb)

Here we track the forward KL divergence value with a Monte Carlo estimate at each update. With the LP-based convexity check active (backtracking by 1/1.25 on infeasibility), the KL decreases steadily over 600 iterations (Fig. 1). From step 60 to 600, it drops from 0.8570 to 0.0438 (about a 95% reduction).

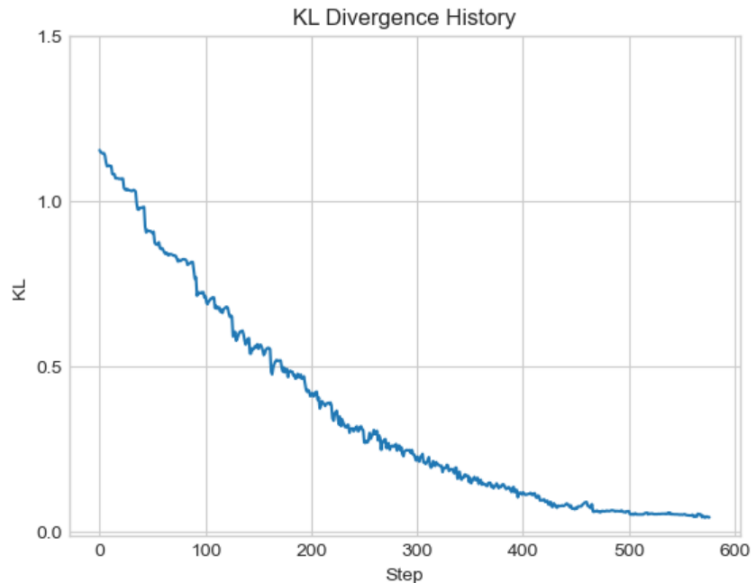


Figure 1: KL divergence per iteration.

2.4 Summary

This modification introduces an adaptive mechanism of β . The LP check is not computationally efficient right now. The main aim of this method is to provide a principled safeguard of global convexity.